COMPUTER ENGINEERING

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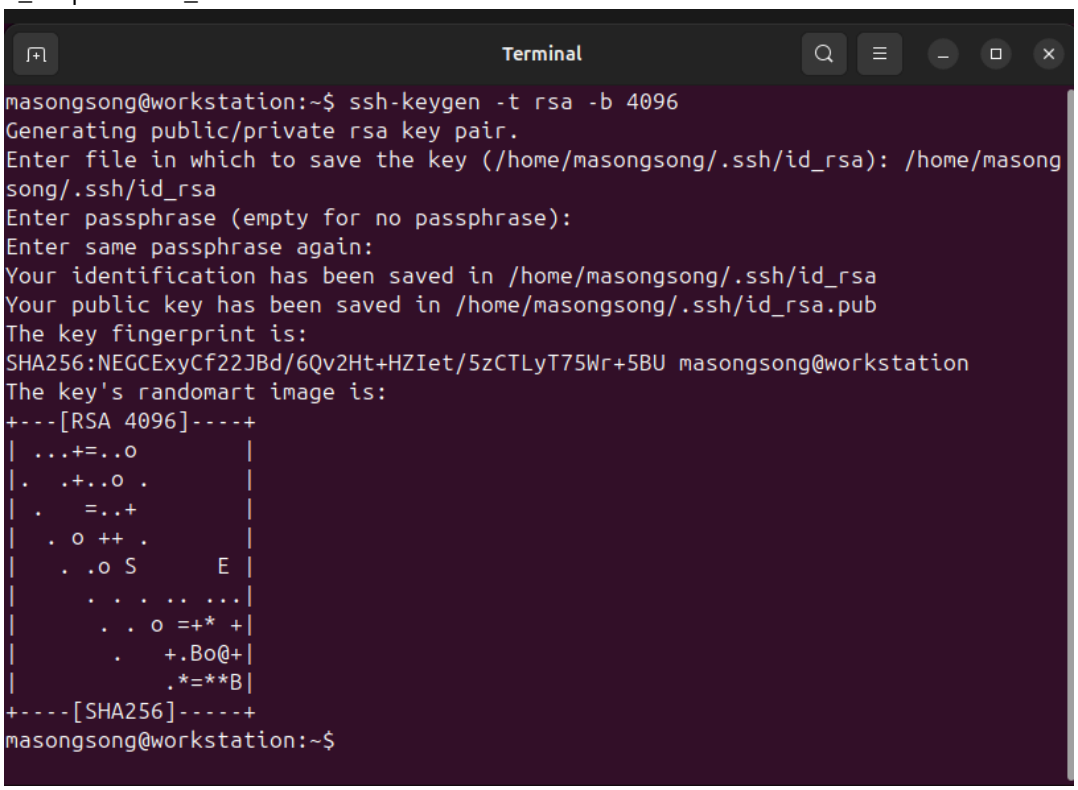
COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	07-16-26	**DATE SUBMITTED:**	07-16-26
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ACTIVITY NO. 2.1 SSH Key-based Authentication and GIT setup

1. PROCEDURE OUTPUT

Task 1

1. The simplest way to generate a key pair is to run ssh-keygen without arguments. In this case, it will prompt for the file in which to store keys. First, the tool asked where to save the file. SSH keys for user authentication are usually stored in the users .ssh directory under the home directory. However, in enterprise environments, the location is often different. The default key file name depends on the algorithm, in this case id_rsa when using the default RSA algorithm. It could also be, for example, id_dsa or id_ecdsa.
2. Issue the command ssh-keygen -t rsa -b 4096. The algorithm is selected using the -t option and key size using the -b option.
3. When asked for a passphrase, just press enter. The passphrase is used for encrypting the key, so that it canno be used even if someone obtains the private key file. The passphrase should be cryptographically strong.
4. Verify that you have created the key by issuing the command ls -la .ssh. The command should show the .ssh directory containing a pair of keys. For example, id_rsa.pub and id_rsa.



Task 2

1. To use public key authentication, the public key must be copied to a server and installed in an authorized_keys file. This can be conveniently done using the ssh-copy-id tool.
2. Issue the command similar to this: ssh-copy-id -i ~/.ssh/id_rsa user@host
3. Once the public key has been configured on the server, the server will allow any connecting user that has the private key to log in. During the login process, the client proves possession of the private key by digitally signing the key exchange.
4. On the local machine, verify that you can SSH with Server 1 and Server 2. What did you notice? Did the connection ask for a password? If not, why?

```
masongsong@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa masongsong@server1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/masongsong/
.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
masongsong@server1's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'masongsong@server1'"
and check to make sure that only the key(s) you wanted were added.

masongsong@workstation:~$
```

```
masongsong@workstation:~$ ssh masongsong@server1
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log
Last login: Thu Jul 16 01:19:23 2026 from 192.168.56.127
masongsong@server1:~$
```

```
masongsong@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa masongsong@server2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/masongsong
.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are promp
ed now it is to install the new keys
masongsong@server2's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'masongsong@server2'"
and check to make sure that only the key(s) you wanted were added.

masongsong@workstation:~$
```

```
masongsong@workstation:~$ ssh masongsong@server2
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log
Last login: Thu Jul 16 01:20:29 2026 from 192.168.56.127
masongsong@server2:~$
```

What did you notice? Did the connection ask for a password? If not, why?

When copying the given command in the terminal it asked me where I wanted it to be placed and gave me a recommended path. After that, it asked for a password to confirm if the keys I added were what I wanted and working. After asking me to login using the keys, it first asked for a password to confirm if I really wanted to login to that user. After the second attempt for the second user, it did not ask me for another password confirmation.

Reflections:
Answer the following:
1. How will you describe the ssh-program? What does it do?
It makes it easier for you to login to different remote servers with only the use of ssh user@host, it utilizes you actually creating a new key pair using keygen and copying the public key for that to the remote servers. Overall it makes it easier to login to different servers as long as you know the keys to it.
2. How do you know that you already installed the public key to the remote servers?
When you try to login using ssh user@host, it just asks for a password and connects you almost immediately to said remote server.

Task 3
1. On the local machine, verify the version of your git using the command which git. If a directory of git is displayed, then you don't need to install git. Otherwise, to install git, use the following command: sudo apt install git

```
masongsong@workstation:~$ sudo apt install git
[sudo] password for masongsong:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb
  git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 1 not upgraded.
Need to get 4,806 kB of archives.
After this operation, 24.5 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 liberror-perl all 0.1
7029-2 [25.6 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 git-man all 1
:2.43.0-1ubuntu7.3 [1,100 kB]
```

2. After the installation, issue the command which git again. The directory of git is usually installed in this location: user/bin/git.

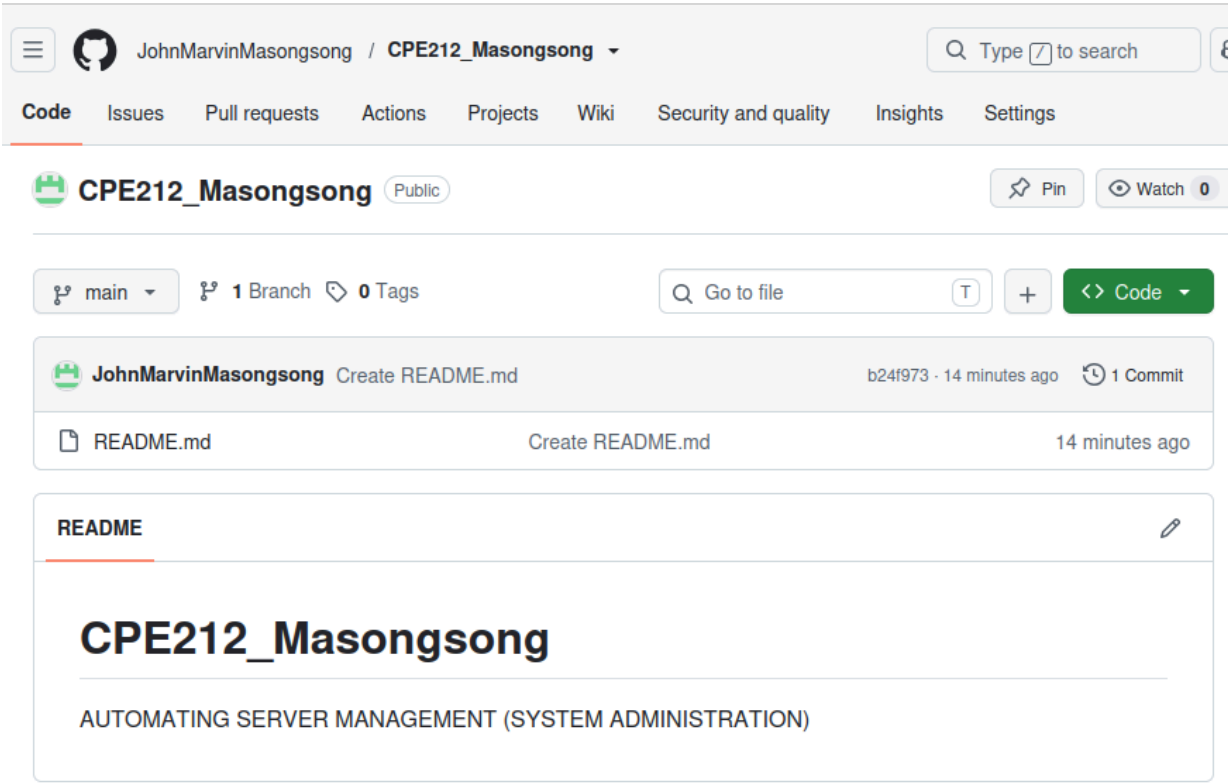
```
masongsong@workstation:~$ which git
/usr/bin/git
masongsong@workstation:~$
```

3. The version of git installed in your device is the latest. Try issuing the command git --version to know the version installed.

```
masongsong@workstation:~$ git --version
git version 2.43.0
masongsong@workstation:~$
```

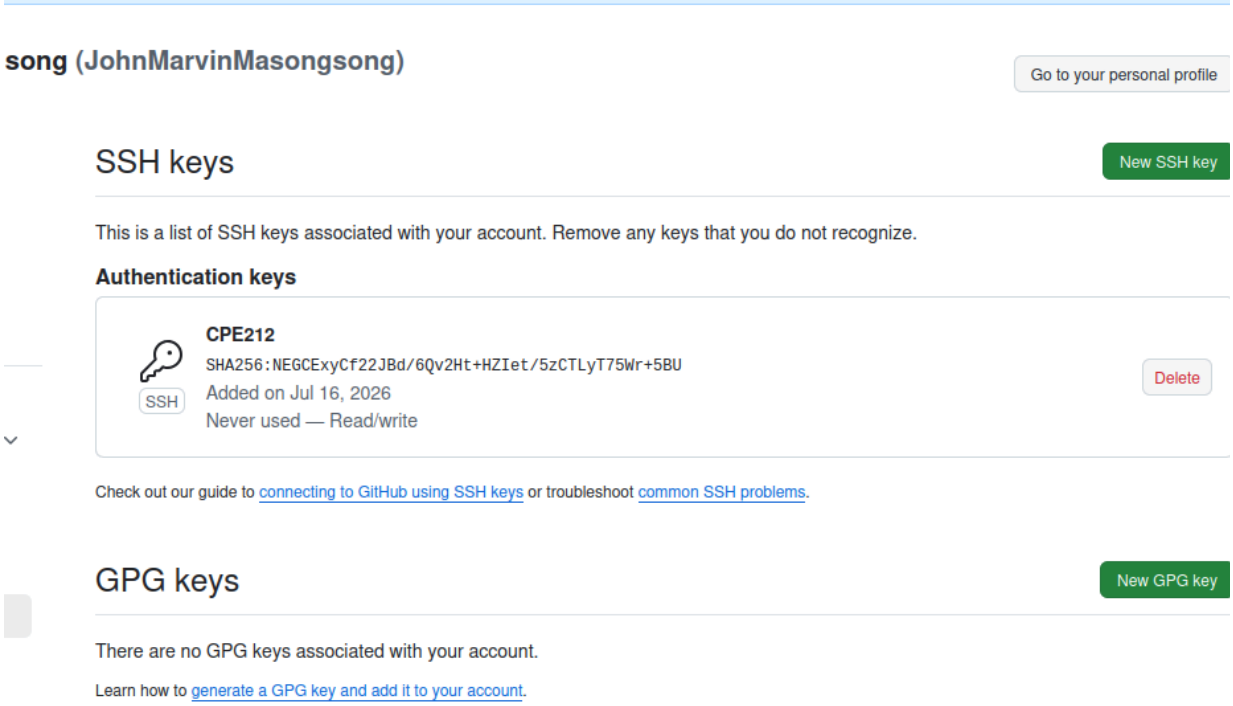
4. Using the browser in the local machine, go to www.github.com.

5. Sign up in case you don't have an account yet. Otherwise, login to your GitHub account.
a. Create a new repository and name it as CPE232_yourname. Check Add a README file and click Create repository.

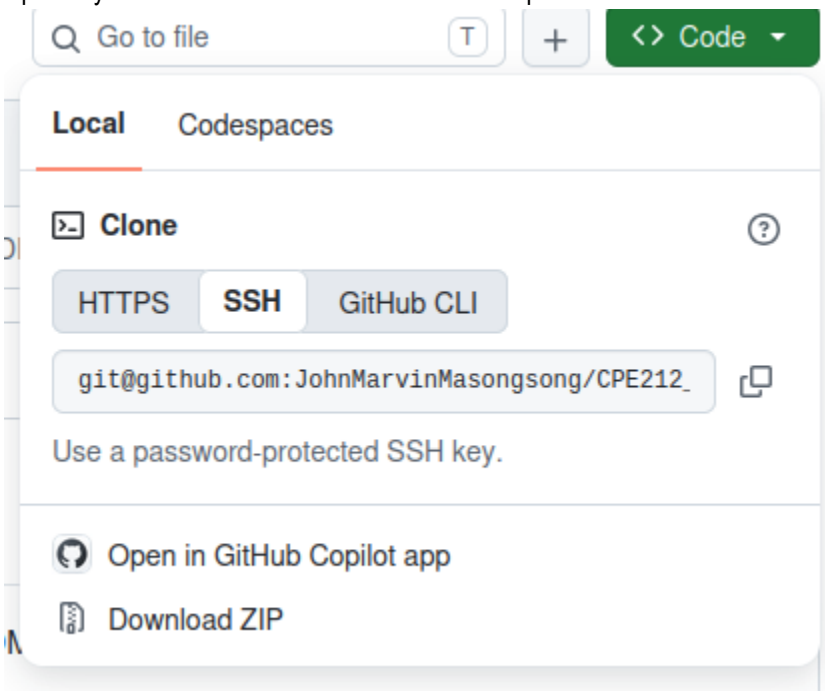


b. Create a new SSH key on GitHub. Go your profile's setting and click SSH and GPG keys. If there is an existing key, make sure to delete it. To create a new SSH keys, click New SSH Key. Write CPE232 key as the title of the key.
c. On the local machine's terminal, issue the command cat .ssh/id_rsa.pub

and copy the public key. Paste it on the GitHub key and press Add SSH key.



d. Clone the repository that you created. In doing this, you need to get the link from GitHub. Browse to your repository as shown below. Click on the Code drop down menu. Select SSH and copy the link.



e.Issue the command git clone followed by the copied link. For example, git clone git@github.com:jvtaylar-cpe/CPE232_yourname.git. When prompted to continue connecting, type yes and press enter.

```
masongsong@workstation:~$ git clone git@github.com:JohnMarvinMasongsong/CPE212_M
asongsong.git
Cloning into 'CPE212_Masongsong'...
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
masongsong@workstation:~$
```

f. To verify that you have cloned the GitHub repository, issue the command ls. Observe that you have the CPE232_yourname in the list of your directories. Use CD command to go to that directory and LS command to see the file README.md.

```
masongsong@workstation:~$ ls
CPE212_Masongsong  Documents  Music     Public  Templates
Desktop            Downloads  Pictures  snap    Videos
masongsong@workstation:~$
```

g. Use the following commands to personalize your git.

• git config --global user.name “Your Name”
• git config --global user.email yourname@email.com
• Verify that you have personalized the config file using the command cat ~/.gitconfig

```
masongsong@workstation:~$ git config --global user.name "Masongsong"
masongsong@workstation:~$ git config --global user.email qjmbmasongsong@tip.edu.
ph
masongsong@workstation:~$ cat ~/.gitconfig
[user]
        name = Masongsong
        email = qjmbmasongsong@tip.edu.ph
masongsong@workstation:~$
```

h. Edit the README.md file using nano command. Provide any information on the markdown file pertaining to the repository you created. Make sure to write out or save the file and exit.

```
masongsong@workstation:~/CPE212_Masongsong$ cat README.md
# CPE212_Masongsong
AUTOMATING SERVER MANAGEMENT (SYSTEM ADMINISTRATION)

COOL STATEMENT FROM MASONG SQUARED (Terminal)
masongsong@workstation:~/CPE212_Masongsong$
```

i. Use the git status command to display the state of the working directory and the staging area. This command shows which changes have been staged, which haven’t, and which files aren’t being tracked by Git. Status output

does not show any information regarding the committed project history. What is the result of issuing this command?

```
masongsong@workstation:~/CPE212_Masongsong$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
masongsong@workstation:~/CPE212_Masongsong$
```

j. Use the command git add README.md to add the file into the staging area.

```
masongsong@workstation:~/CPE212_Masongsong$ git add README.md
masongsong@workstation:~/CPE212_Masongsong$
```

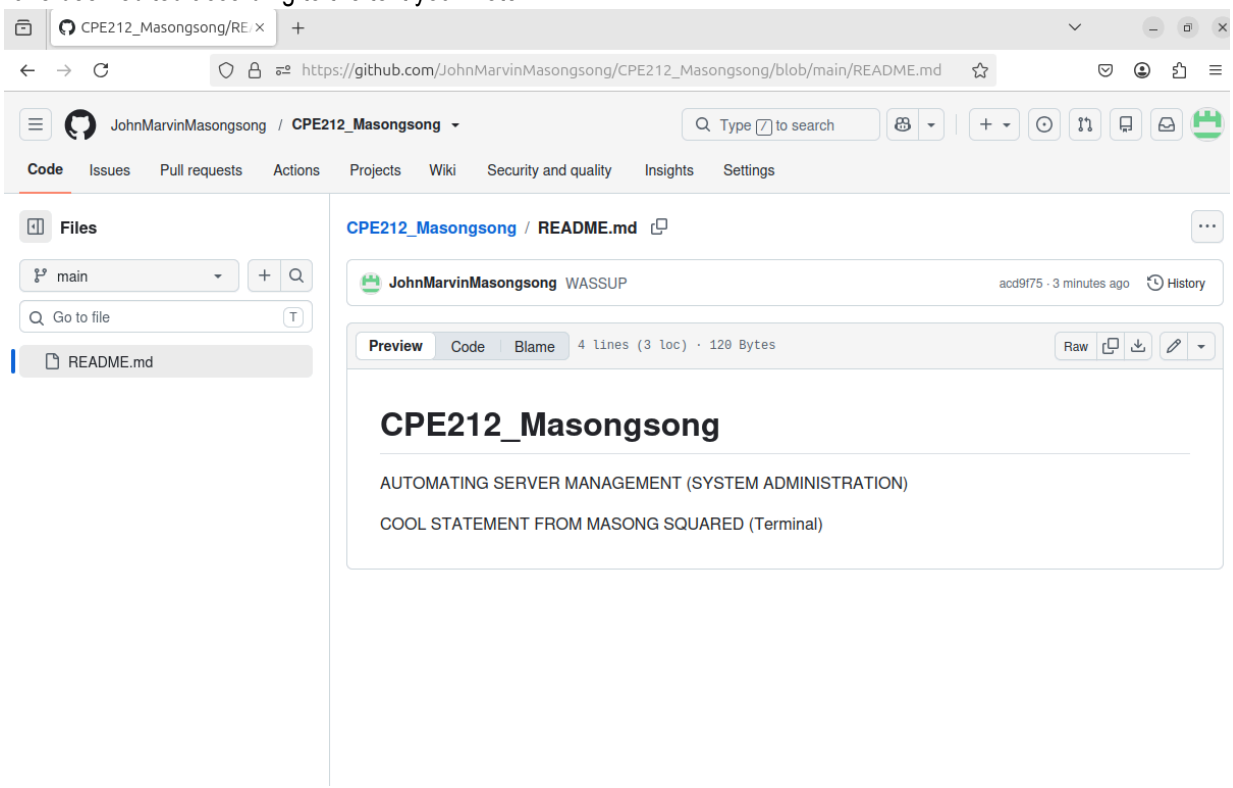
k. Use the git commit -m “your message” to create a snapshot of the staged changes along the timeline of the Git projects history. The use of this command is required to select the changes that will be staged for the next Commit.

```
masongsong@workstation:~/CPE212_Masongsong$ git commit -m "WASSUP"
[main acd9f75] WASSUP
 1 file changed, 2 insertions(+)
masongsong@workstation:~/CPE212_Masongsong$
```

l. Use the command git push <remote><branch> to upload the local repository content to GitHub repository. Pushing means to transfer commits from the local repository to the remote repository. As an example, you may issue git push origin main

```
masongsong@workstation:~/CPE212_Masongsong$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 352 bytes | 352.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:JohnMarvinMasongsong/CPE212_Masongsong.git
   b24f973..acd9f75  main -> main
masongsong@workstation:~/CPE212_Masongsong$
```

m. On the GitHub repository, verify that the changes have been made to README.md by refreshing the page. Describe the README.md file. You can notice the how long was the last commit. It should be some minutes ago and the message you typed on the git commit command should be there. Also, the README.md file should have been edited according to the text you wrote.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. What sort of things have we so far done to the remote servers using ansible commands?
   - We have done the connection from different remote servers checking if they are pingable and reachable via another remote machine.

4. How important is the inventory file?
   - The inventory file is important because it is able to store the servers that ansible will be managing, connecting, and pinging.

Conclusion:

I have learned a lot from this HOA, mainly the use of previous knowledge into SSH and with the use of ansible. We also learned how to setup git in linux and how to manage it from the terminal. It was quite hard to understand the statements, but once I saw the picture I was able to do the tasks at hand. All in all, the knowledge they gave me allowed me to progress and I hope to use this in my future HOAs.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **NA**

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **NA**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **NA**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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